CNN Model Improvement and Optimization Report

This report summarizes the progressive improvements applied to the Convolutional Neural Network (CNN) model developed for food delivery time prediction. The project initially started with a basic CNN implementation using synthetic spatial route images generated from structured delivery data. Multiple enhancement strategies were systematically applied to improve learning capability, stability, validation performance, and generalization.

1. Initial CNN Performance Challenges

The initial CNN model struggled to achieve strong predictive performance because the dataset consisted of structured tabular delivery information rather than natural image data. The first synthetic route images were overly simplistic and contained limited spatial complexity. As a result: Initial accuracy remained close to random prediction levels. The CNN could not extract rich visual patterns. Cross-validation results showed unstable generalization. These observations motivated a series of architectural, preprocessing, and representation improvements.

2. Step-by-Step Improvements Applied

Improvement	Purpose	Impact
Haversine Distance Feature	Improved geographical realism	Enhanced route-based spatial information
Weather Feature Engineering	Added environmental context	Introduced additional delivery-condition patterns
Synthetic Spatial Image Enhancement	Added richer visual structures	Improved CNN feature extraction capability
Traffic-Based Route Coloring	Encoded traffic intensity visually	Improved route representation diversity
Background Texture Variations	Simulated weather conditions visually	Added spatial complexity
Batch Normalization	Stabilized CNN training	Reduced internal covariate shift
Dropout Regularization	Reduced overfitting risk	Improved generalization
GlobalAveragePooling2D	Reduced parameter explosion	Improved model efficiency
Data Augmentation	Increased training variability	Improved robustness
5-Fold Cross-Validation	Improved evaluation reliability	Measured generalization stability
RandomizedSearchCV	Optimized hyperparameters	Improved validation accuracy

3. CNN Architecture Improvements

The CNN architecture evolved considerably during experimentation. The initial model used standard convolutional and pooling layers followed by Flatten and dense layers. However, this produced excessive parameters and unstable learning behavior. To address these issues: GlobalAveragePooling2D replaced Flatten layers to reduce parameter count. BatchNormalization layers improved training stability. Dropout layers reduced overfitting. Filter sizes and kernel sizes were optimized. Learning rates were adjusted for smoother convergence. These improvements produced more stable training and better validation performance.

4. Hyperparameter Tuning Results

Hyperparameter tuning was performed using RandomizedSearchCV with 5-fold cross-validation. Multiple combinations of filters, kernel sizes, learning rates, epochs, and batch sizes were explored. The best-performing configuration was: Filters = 32 Kernel Size = 3×3 Learning Rate = 0.0001 Epochs = 15 Batch Size = 16 The tuned

CNN achieved a best cross-validation accuracy of approximately 56.88%, representing a noticeable improvement over the initial CNN implementation.

5. Learning Curve and Validation Analysis

Learning curves showed that training and validation accuracy remained relatively close across epochs. This indicated that the model did not suffer from severe overfitting. However: Validation accuracy fluctuated across epochs. Validation loss occasionally spiked. K-fold accuracies varied between folds. These behaviors suggested: Mild underfitting Moderate generalization instability Dataset limitations and synthetic image simplification Despite these challenges, augmentation, regularization, and tuning improved stability significantly compared to the original model.

6. Final Model Performance

Metric	CNN	Logistic Regression
Accuracy	52.5%	62.5%
Precision	52.9%	64.7%
Recall	45%	55%
F1 Score	0.486	0.595
ROC AUC	0.58	0.74

Although Logistic Regression ultimately outperformed CNN on the structured delivery dataset, the CNN demonstrated substantial improvement after multiple enhancement strategies were applied. The experimentation process successfully demonstrated advanced CNN optimization workflows, validation strategies, and performance analysis techniques.

7. Final Conclusion

This project demonstrated a complete deep learning experimentation workflow involving preprocessing, feature engineering, image generation, CNN architecture optimization, augmentation, cross-validation, hyperparameter tuning, and comparative evaluation. The history of improvements showed that: Richer synthetic image representations improved CNN learning capability. Regularization and architectural redesign improved stability. Cross-validation provided realistic generalization estimates. Hyperparameter tuning enhanced validation performance. Algorithm suitability is highly dependent on data type. The project successfully highlighted both the strengths and limitations of applying CNNs to structured delivery prediction problems using synthetic spatial imagery.